# Steady-State Thermal Modelling of a Photovoltaic–Thermal Panel

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## Abstract

This report presents a steady-state thermal model developed to estimate temperature distributions and heat exchanges within a photovoltaic–thermal (PVT) panel. The panel is represented by a sequence of solid layers and a simplified water layer, connected through conductive, convective and radiative exchanges. A system of eleven nonlinear energy balances provides the temperature states, while an auxiliary electrical calculation estimates the PV output. The reported case gives upper-layer temperatures of approximately 385 K and an electrical output of 272.4 W. The results provide an initial numerical description of the panel, with limitations associated with the flow assumptions, boundary temperatures and coupling between thermal and electrical calculations. Experimental comparison and further refinement remain necessary.

## 1 Introduction and objective

Photovoltaic panels convert incident solar radiation into electricity. Photovoltaic–thermal collectors combine this function with heat recovery, allowing part of the thermal energy absorbed by the panel to be transferred to a circulating fluid. The temperatures of the PV cells and surrounding layers influence both the heat available for recovery and the electrical performance of the system.

The objective of this work was to develop a reduced steady-state model of a PVT panel and use it to calculate the temperatures and heat exchanges associated with a specified operating case. The formulation follows the layer-based description and selected correlations discussed by Herrando et al. [1], together with heat-transfer relations from the Technical Physics course [2]. The reduced formulation is used here to examine the thermal behaviour of the panel under the adopted operating conditions.

The report first defines the physical representation and assumptions, then presents the governing relations and input data. The calculated temperatures and heat-transfer quantities are discussed before the limitations and proposed developments. Appendix A provides the complete set of 94 numerical entries associated with the calculation.

## 2 Physical representation and assumptions

The panel is represented as a flat, rectangular arrangement of glass, EVA encapsulant, PV cells, Tedlar, an intermediate layer, an absorber assembly and back insulation. The water layer is also represented as flat and rectangular. This representation supports a one-dimensional description of the exchanges between neighbouring temperature nodes, rather than resolving the detailed geometry of the flow channels.

Steady state is assumed, so energy storage and temporal variations are excluded from the balances. Ambient conditions are treated as uniform, frame heat losses are neglected, and the water and external-air flows are simplified as one-dimensional. Radiation absorption in the glass and collector frame is treated as negligible, while an irradiance correction of 50 W/m² is included in the adopted input set.

Each layer or heat-transfer mechanism introduces a resistance or exchange relation between temperature nodes. Figure 1 shows the thermal resistance network used in the original work. The symbols $T_1$–$T_{11}$ refer to the temperature states in the equation set; external and fluid temperatures are identified separately.

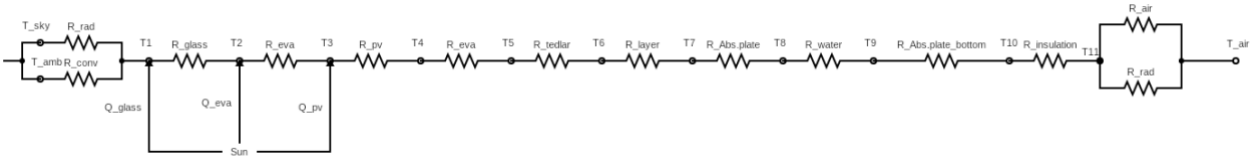



**Figure 1. Thermal resistance network reproduced from the final source report.**

# 3 Mathematical formulation

## 3.1 Convection and thermal resistances

The Richardson number was used to assess the relative importance of natural and forced convection. The criteria adopted in the source are Ri < 0.1 when natural convection is negligible, Ri > 10 when forced convection is negligible, and 0.1 < Ri < 10 when both contributions matter. The model uses forced-convection correlations for the reported conditions.

$$Ri = \frac{Gr}{\mathrm{Re}^2} \tag{1}$$

The wind-side heat-transfer coefficient is evaluated from the empirical relation used in [1]. The source also denotes this coefficient by $h_{12}$method and uses Vair for the prescribed wind speed.

$$h_{wind} = 4.5 + 2.9 * (V_{wind}) \tag{2}$$

The upper- and lower-air film temperatures are defined by the following mean-temperature relations. They provide the temperature arguments associated with the transport-property evaluation in the adopted formulation.

$$T_{film_1} = \frac{T_1 + T_{amb}}{2} \tag{3}$$

$$T_{film_3} = \frac{20 + 273{,}15 \ + T_{11}}{2} \tag{4}$$

Reynolds and Prandtl numbers are used in the adopted Nusselt correlations to obtain the water-side and lower-air heat-transfer coefficients. The following expressions define the Reynolds and Nusselt relations used in the calculation. The viscosity parameters retain the implementation names v and vwater.

$$\mathrm{Re1} \ = \frac{ro_{air} * V_{air} * S_{glass}}{v} \tag{5}$$

$$\mathrm{Re2} = \frac{ro_w * V_{avg} * Dh}{v_{water}} \tag{6}$$

$$\mathrm{Re3} \ = \frac{ro_{air} * V_{air} * S_{ins}}{v} \tag{7}$$

$$Nu1 \ = \ 0{,}906 * \mathrm{Re1}^{0{,}5} * Pr1^{\frac{1}{3}} \tag{8}$$

$$Nu = \ \ 0{,}023 * \mathrm{Re2}^{0{,}8} * Pr2^{\frac{1}{3}} \tag{9}$$

$$Nu3 = 0{,}906 * \mathrm{Re}3^{0,5} * Pr3^{\frac{1}{3}} \tag{10}$$

The water-side Nusselt relation is written as Nu below and corresponds to $Nu_2$ in the numerical listing. The coefficients $h_2$ and $h_3$ are then obtained from the following relations, where $\lambda_1$ and $\lambda_2$ are the fluid conductivity parameters.

$$h_2 = \frac{Nu2 * lambda2}{U_2} \tag{11}$$

$$h_3 = \frac{Nu3 * lambda1}{L1} \tag{12}$$

For a solid layer with thickness S and conductivity K, the conductive resistance follows the relation R = S/K. In this one-dimensional, area-normalised representation, the resistance has units of m² K/W. Material values and thicknesses are listed in Table 2.

## 3.2 Optical, electrical and radiative relations

The short-wave transmittance of the glass determines how much incident radiation reaches the PV layer, while the PV absorption coefficient determines the absorbed contribution. The adopted glass transmittance and PV absorption coefficient are 0.94 and 0.95, respectively. The expression used for the PV irradiance is [1]:

$$G_{pv} = G * t\mathrm{glass}_{\mathrm{short}} * \alpha_{pv_{\mathrm{short}}} \tag{13}$$

The electrical power per unit area and temperature-dependent efficiency are described by the relations below. The reference efficiency ηPV,ref is specified at the reference cell temperature TPV,ref = 25 °C. The coefficient $\beta_o$ describes the temperature dependence of electrical power [1].

$$W_e = G \cdot \tau g, short \cdot \eta PV \tag{14}$$

$$\eta PV = \eta PV, ref \cdot [1 + \beta 0 \cdot (T_{PV} - T_{PV}, ref)] \tag{15}$$

In the saved auxiliary calculation, the nominal efficiency is 18.44%, the adopted coefficient is $B_o$ = −0.4%/K, and the cell-temperature argument is fixed at 35 °C. This gives the reported efficiency of 17.7%. The corresponding total absorbed radiation and electrical output are calculated as follows:

$$eta_{pv} = eta_{pv_{ref}} * \left(1 + \left(\frac{Bo}{100}\right) * \left((35 + 273{,}15) - T_{pv_{ref}}\right)\right) \tag{16}$$

$$Qrad = G_p * \cos(teta) * A * a_{pv_{short}} \tag{17}$$

$$Pe = Qrad * \left(\frac{eta_{pv}}{100}\right) \tag{18}$$

EVA is treated as a highly transmitting encapsulant, and its separate contribution to the final optical attenuation is neglected. Radiative exchanges from the glass and PV surfaces to the sky are represented by the two expressions retained from the source model:

$$Q_{radsky1} = e_{glass_{long}} * k * \left((T_1^4) - \left(T_{sky}^4\right)\right) \tag{19}$$

$$Q_{radsky2} = e_{pv_{long}} * k * \left((T_3^4) - \left(T_{sky}^4\right)\right) \tag{20}$$

The sky temperature is estimated from ambient temperature using the relation adopted from [1]. Temperatures in this expression are in kelvin. The reported case uses an ambient temperature of 20 °C and gives a sky temperature of approximately 4 °C.

$$T_{sky} = \ 0{,}0552 * (T_{amb})^{1{,}5} \tag{21}$$

## 3.3 Coupled layer energy balances

Eleven nonlinear balances relate the temperature states to the heat entering and leaving successive layers. They combine conduction between layers with convective and radiative exchanges at the relevant interfaces. The node indices identify the temperatures and interfaces shown in Figure 1.

$$G_p + \left(\frac{T_2 - T_1}{R_{glass}}\right) - \ h1_{2_{method}} * \left(T_{film_1} - T_{amb}\right) - \ e_{glass_{long}} * k * \left((T_1^4) - \left(T_{sky}^4\right)\right) = \ 0 \tag{22}$$

$$G_{pv_2} + \left(\frac{T_3 - T_2}{R_{Eva}}\right) - \left(\frac{T_2 - T_1}{R_{glass}}\right) = 0 \tag{23}$$

$$Gpv \ + \ Pe \ - \left(\frac{T_3 - T_2}{R_{Eva}}\right) - \left(\frac{T_4 - T_5}{R_{Eva}}\right) - \ e_{pv_{long}} * k * \left((T_3^4) - \left(T_{sky}^4\right)\right) = \ 0 \tag{24}$$

$$\left(\frac{T_4 - T_5}{R_{Eva}}\right) - \left(\frac{T_5 - T_6}{R_{Tedlar}}\right) = \ 0 \tag{25}$$

$$\left(\frac{T_5 - T_6}{R_{Tedlar}}\right) - \left(\frac{T_6 - T_7}{R_{layer}}\right) = \ 0 \tag{26}$$

$$\left(\frac{T_6 - T_7}{R_{layer}}\right) - \left(\frac{T_7 - T_8}{R_{abs}}\right) = \ 0 \tag{27}$$

$$\left(\frac{T_7 - T_8}{R_{abs}}\right) - \ h_2 * \left(T_8 - \frac{T_8 + T_{water}}{2}\right) = \ 0 \tag{28}$$

$$h_2 * \left(T_8 - \frac{T_8 + T_{water}}{2}\right) - \ h_2 * \left(\left(\frac{T_{water} + T_8}{2}\right) - T_9\right) = \ 0 \tag{29}$$

$$h_2 * \left(\left(\frac{T_{water} + T_8}{2}\right) - T_9\right) - \left(\frac{T_9 - T_{10}}{R_{absB}}\right) = \ 0 \tag{30}$$

$$\left(\frac{T_9 - T_{10}}{R_{absB}}\right) - \left(\frac{T_{10} - T_{11}}{R_{ins}}\right) = 0 \tag{31}$$

$$\left(\frac{T_{10} - T_{11}}{R_{ins}}\right) - \ h_3 * (T_{11} - T_{air}) = \ 0 \tag{32}$$

Equations (22)–(32) describe, in sequence, the glass-side exchange, the glass/EVA connection, the PV-related balance, conduction through the EVA/Tedlar and intermediate layers, transfer to the absorber, exchanges across the water region, conduction through the lower absorber and insulation, and exchange with the lower-air boundary. The coupled system provides the layer temperature states discussed in Section 5.

# 4 Operating conditions and model data

The operating case is defined by the imposed irradiance, ambient and water temperatures, fluid parameters, geometry and electrical reference properties. Table 1 collects the principal operating quantities. The reference conditions and transport parameters define the numerical case considered here.

**Table 1. Principal operating and reference quantities used in the reported calculation.**

| Quantity / source symbol | Value | Unit / definition |
|---|---|---|
| Incident irradiance, Gp | 1000 | W/m² |
| Irradiance parameter, Gpv2 | 950 | W/m² |
| Ambient temperature, Tamb | 20 | °C; reported output 293.2 K |
| Water temperature, Twater | 25 | °C; reported output 298.2 K |
| Wind speed, Vair | 10 | m/s |
| Gross area, A | 1.62 | m² |
| Reference PV efficiency, etapvref | 18.44 | % |
| Power temperature coefficient, Bo | -0.4 | %/K |
| Reference PV temperature, Tpvref | 25 | °C |
| Electrical temperature argument | 35 | °C, fixed in the saved calculation |
| Incidence angle, teta | 0 | normal incidence |
| Water mass flow, mwater | 0.3155 | kg/s |
| Water average speed, Vavg | 0.06309 | m/s |
| Hydraulic diameter, Dh | 0.03846 | m |

The water mass-flow rate is obtained from the nominal density parameter dens and the volumetric-flow parameter Velo. Their product gives the adopted mass flow of 0.3155 kg/s:

$$m_{water} = dens * Velo \tag{33}$$

Table 2 groups the solid-layer properties. The conductivities are expressed in W/(m K), consistent with the material-property definition in [1], and the thicknesses in metres. The adopted glass conductivity is 14 W/(m K).

**Table 2. Layer thicknesses and conductivity parameters retained from the source model.**

| Layer | Thickness S (m) | Conductivity K (W/(m K)) |
|---|---|---|
| Glass | 0.0032 | 14 |
| PV cells | 0.00035 | 130 |
| EVA | 0.0005 | 0.35 |
| Tedlar | 0.0003 | 0.36 |
| Intermediate layer | 0.00035 | 0.35 |
| Upper absorber | 0.0015 | 310 |
| Lower absorber | 0.0015 | 310 |
| Insulation | 0.03 | 0.034 |

The adopted optical properties are given in Table 3. The symbols apvshort and alphapvshort are retained in Appendix A because both occur in the source output, although they have the same reported value.

**Table 3. Optical properties used in the model.**

| Property / source symbol | Value |
|---|---|
| Glass short-wave transmittance, tglassshort | 0.94 |
| Glass long-wave transmittance, tglasslong | 0.06 |
| Glass short-wave emissivity, eglassshort | 0.05 |
| Glass long-wave emissivity, eglasslong | 0.86 |
| PV long-wave emissivity, epvlong | 0.89 |
| PV short-wave absorption, alphapvshort | 0.95 |

The remaining fluid, geometric and transport parameters and the calculated coefficients are listed in Appendix A. The implementation variables are kept distinct so that each value can be associated with its role in the model.

# 5 Results and discussion

## 5.1 Temperature distribution

Table 4 reports the eleven temperature states. The upper states $T_1$, $T_2$ and $T_3$ range from 384.6 to 385.4 K, while the central states $T_5$–$T_8$ are reported as 298 K. The lower states $T_9$ and $T_{10}$ are 298.2 K, and $T_{11}$ is 310.7 K. These values are outputs of the simplified model and provide the temperature pattern to be assessed against the formulation and boundary conditions.

**Table 4. Reported layer temperature states; values are retained at the source precision.**

| Temperature state | Reported value (K) |
|---|---|
| T1 | 384.6 |
| T2 | 384.9 |
| T3 | 385.4 |
| T4 | 297.9 |
| T5 | 298 |
| T6 | 298 |
| T7 | 298 |
| T8 | 298 |
| T9 | 298.2 |
| T10 | 298.2 |
| T11 | 310.7 |

The reported lower-air temperature Tair is 317.3 K, higher than $T_{11}$. The resulting temperature ordering in the lower region requires further investigation of the boundary-temperature definition and heat-transfer assumptions.

## 5.2 Electrical output and heat-transfer quantities

The auxiliary electrical calculation gives an efficiency of 17.7% and a total electrical output of 272.4 W. The reported water-side heat-transfer quantity is 1532 W, with an outlet-water temperature of 299.3 K. Table 5 brings together the principal reported outputs; the complete numerical listing is retained in Appendix A.

**Table 5. Selected performance quantities from the reported calculation.**

| Quantity / source symbol | Value | Unit |
|---|---|---|
| PV efficiency, etapv | 17.7 | % |
| Electrical output, Pe | 272.4 | W |
| Absorbed-radiation quantity, Qrad | 1539 | W |
| Water heat-transfer quantity, Qwater | 1532 | W |
| Outlet-water temperature, Twaterout | 299.3 | K |
| Wind-side coefficient, h12method | 33.5 | W/(m² K) |
| Water-side coefficient, h2 | 175.1 | W/(m² K) |
| Lower-air coefficient, h3 | 2.141 | W/(m² K) |

The electrical calculation uses a fixed cell-temperature argument of 35 °C. This imposed quantity is distinct from the temperature states obtained from the layer balances; the implications for thermal–electrical coupling are discussed in Section 6.

# 6 Limitations and further development

The main limitations arise from the simplified geometry, the assumed flow regimes and the treatment of the boundary temperatures. The one-dimensional water representation does not resolve the channel flow distribution or demonstrate uniform heat absorption. A subsequent CFD model could examine these effects in greater detail, after the reduced formulation and input conventions have been checked.

The transport-property conventions require a consistent definition before further quantitative assessment. In particular, the viscosity labels and units, the physical interpretation of $U_2$, and the separate water-density parameters should be reconciled with the Reynolds and heat-transfer relations. The adopted correlations also need to be checked against their applicable flow conditions. The electrical calculation currently uses a prescribed temperature rather than feedback from the calculated PV-layer temperature.

The model results remain preliminary. A numerical review should verify the energy-balance signs, dimensional consistency, boundary definitions and coupling between the electrical and thermal calculations. Comparison with measured panel temperatures and heat outputs is then required to assess predictive accuracy. No experimental validation of this model is claimed in the present report.

# 7 Conclusions

The study established a layer-based steady-state representation of a PVT panel and produced a set of temperature and heat-transfer outputs for the adopted operating case. The combined temperature tables, equation set and parameter listing make the numerical formulation and its results available for further inspection and development.

The reported temperature distribution and electrical output illustrate the behaviour of the simplified calculation, while the lower-region result and the unresolved input conventions define the main priorities for refinement. Further work should first consolidate the model definitions and then extend the assessment through experimental comparison and, where useful, a more detailed CFD representation.

# References

[1] M. Herrando, G. Fantoni, A. Cubero, R. Simón-Allué, I. Guedea and N. Fueyo, “Numerical analysis of the fluid flow and heat transfer of a hybrid PV-thermal collector and performance assessment,” Renewable Energy, vol. 209, pp. 122–132, June 2023.

[2] University of Tuscia, Department of Engineering, Technical Physics course materials.

[3] Engineering ToolBox, “Water – Dynamic and Kinematic Viscosity,” 2019. https://www.engineeringtoolbox.com/water-dynamic-kinematic-viscosity-d_596.html

# Appendix A Complete reported numerical listing

Table A.1 lists all 94 numerical entries associated with the calculation. Implementation variable names are retained to support direct comparison with the model. Conductivities, area-normalised resistances, specific heat and the radiation coefficient are expressed in SI units. The viscosity and $U_2$ conventions requiring clarification are identified in Section 6.

**Table A.1. Complete numerical listing for the calculated operating case.**

| Source variable | Reported value | Unit / convention |
|---|---|---|
| A | 1.62 | m² |
| alphapvshort | 0.95 | – |
| apvshort | 0.95 | – |
| Bo | -0.4 | %/K |
| cp | 4186 | J/(kg K) |
| dens | 1000 | kg/m³ |
| Dh | 0.03846 | m |
| etapv | 17.7 | % |
| etapvref | 18.44 | % |
| eglasslong | 0.86 | – |
| eglassshort | 0.05 | – |
| epvlong | 0.89 | – |
| Gpv | 893 | W/m² |
| Gp | 1000 | W/m² |
| Gpv2 | 950 | W/m² |
| h12method | 33.5 | W/(m² K) |
| h2 | 175.1 | W/(m² K) |
| h3 | 2.141 | W/(m² K) |
| k | 5.675E-08 | W/(m² K⁴) |
| Kabs | 310 | W/(m K) |
| KabsB | 310 | W/(m K) |
| Keva | 0.35 | W/(m K) |
| Kglass | 14 | W/(m K) |
| Kins | 0.034 | W/(m K) |
| Klayer | 0.35 | W/(m K) |
| Kpv | 130 | W/(m K) |
| Ktedlar | 0.36 | W/(m K) |
| L1 | 1.5 | m |
| lambda1 | 0.0261 | W/(m K) |
| lambda2 | 0.637 | W/(m K) |
| mwater | 0.3155 | kg/s |
| Nu1 | 40.19 | – |
| Nu2 | 27.49 | – |
| Nu3 | 123.1 | – |
| Pe | 272.4 | W |

| Source variable | Reported value | Unit / convention |
|---|---|---|
| Pr1 | 0.73 | – |
| Pr2 | 3.79 | – |
| Pr3 | 0.73 | – |
| Qrad | 1539 | W |
| Qcond111 | -1760 | W |
| Qconvext | 1532 | W |
| Qconvext2 | -14.19 | W |
| Qradsky1 | 780.4 | W |
| Qradsky2 | 817.1 | W |
| Qradsun | 1539 | W |
| Qwater | 1532 | W |
| Re1 | 2428 | – |
| Re2 | 4034 | – |
| Re3 | 22759 | – |
| roair | 1.185 | kg/m³ |
| row | 889.4 | kg/m³ |
| Rabs | 0.000004839 | m² K/W |
| RabsB | 0.000004839 | m² K/W |
| REva | 0.001429 | m² K/W |
| Rglass | 0.0002286 | m² K/W |
| Rins | 0.8824 | m² K/W |
| Rlayer | 0.001 | m² K/W |
| RTedlar | 0.0008333 | m² K/W |
| Sabs | 0.0015 | m |
| SabsB | 0.0015 | m |
| Seva | 0.0005 | m |
| Sglass | 0.0032 | m |
| Sins | 0.03 | m |
| Slayer | 0.00035 | m |
| Spv | 0.00035 | m |
| Stedlar | 0.0003 | m |
| teta | 0 | – |
| T1 | 384.6 | K |
| T10 | 298.2 | K |
| T11 | 310.7 | K |
| T2 | 384.9 | K |
| T3 | 385.4 | K |
| T4 | 297.9 | K |
| T5 | 298 | K |
| T6 | 298 | K |
| T7 | 298 | K |
| T8 | 298 | K |

| Source variable | Reported value | Unit / convention |
|---|---|---|
| T9 | 298.2 | K |
| Tair | 317.3 | K |
| Tamb | 293.2 | K |
| Tfilm1 | 338.9 | K |
| Tfilm3 | 301.9 | K |
| tglasslong | 0.06 | – |
| tglassshort | 0.94 | – |
| Tpvref | 298.2 | K |
| Tsky | 277.1 | K |
| Twater | 298.2 | K |
| Twaterout | 299.3 | K |
| U2 | 0.1 | m in source; see Section 6 |
| v | 0.00001562 | source convention; see Section 6 |
| Velo | 0.0003155 | $m^3/s$ |
| Vair | 10 | m/s |
| Vavg | 0.06309 | m/s |
| vwater | 0.000535 | source convention; see Section 6 |